

Habitat fragmentation mediates the mechanisms underlying long-term climate-driven thermophilization in birds

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Abstract

Climatic warming can shift community composition driven by the colonization-extinction dynamics of species with different thermal preferences; but simultaneously, habitat fragmentation can mediate species' responses to warming. As this potential interactive effect has proven difficult to test empirically, we collected data on birds over 10 years of climate warming in a reservoir subtropical island system that was formed 65 years ago. We investigated how the mechanisms underlying climate-driven directional change in community composition were mediated by habitat fragmentation. We found thermophilization driven by increasing warm-adapted species and decreasing cold-adapted species in terms of trends in colonization rate, extinction rate, occupancy rate and population size. Critically, colonization rates of warm-adapted species increased faster temporally on smaller or less isolated islands; cold-adapted species generally were lost more quickly temporally on closer islands. This provides support for dispersal limitation and microclimate buffering as primary proxies by which habitat fragmentation mediates species range shift. Overall, this study advances our understanding of biodiversity responses to interacting global change drivers.

eLife assessment

This **important** study enhances our understanding of how habitat fragmentation and climate change jointly influence bird community thermophilization in a fragmented island system. The evidence supporting some conclusions is **incomplete**, as while the overall trends are **convincing**, some methodological aspects, particularly the isolation metrics and interpretation of colonization/extinction rates, require further clarification. This work will be of broad interest to ecologists and conservation biologists, providing crucial insights into how ecosystems and communities react to climate change.

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1 Introduction

One widespread consequence of anthropogenic-caused climate warming is the shift in species' distributions toward cooler regions (Chen et al., 2011 [↗](#); La Sorte & Thompson, 2007 [↗](#); Thomas & Lennon, 1999 [↗](#)). As species move across the globe, the composition of community changes, with species both colonizing and departing (Menéndez et al., 2006 [↗](#); Tingley & Beissinger, 2013 [↗](#)); yet colonizing and departing species are unlikely to be randomly drawn from species pools, but instead will be drawn depending on their traits, such as being warm- adapted versus cold-adapted. As a result, a warming climate should lead to compositional shifts toward relatively more warm-adapted species in local communities (Thomas et al., 2006 [↗](#)), a phenomenon termed “thermophilization” (Feeley et al., 2020 [↗](#); Gottfried et al., 2012 [↗](#); Zellweger et al., 2020 [↗](#)).

In practice, thermophilization can be measured by a temporal increase in the Community Temperature Index (CTI) (Devictor et al., 2008 [↗](#)) – an estimation of the average temperature experienced by all species in an assemblage throughout their respective ranges. While thermophilization of communities is usually accompanied by an increase in warm- adapted species and a decrease in cold-adapted species (Curley et al., 2022 [↗](#); Prince & Zuckerberg, 2015 [↗](#); Tayleur et al., 2016 [↗](#)) (Figure 1a [↗](#)), this process is usually spatially heterogeneous and depends on the availability of current habitat and the ability to disperse into newly suitable habitats (Gaüzère et al., 2017 [↗](#); Richard et al., 2021), both of which will be directly or indirectly affected by other global change stressors. Habitat fragmentation (Fahrig, 2003 [↗](#)), in particular, has been hypothesized to have interactive effects with climate change on community dynamics. Specifically, habitat fragmentation usually alters both habitat spatial structure and microclimate characteristics (Ewers & Banks-Leite, 2013 [↗](#); Fahrig, 2003 [↗](#)), either of which can influence thermophilization processes (Gaüzère et al., 2017 [↗](#); Platts et al., 2019 [↗](#); Richard et al., 2021; Zellweger et al., 2020 [↗](#)). Understanding the interaction between climate change and habitat fragmentation on community composition is essential to predicting biodiversity outcomes but we generally lack empirical examples with which to study this phenomenon (Opdam & Wascher, 2004 [↗](#)).

Theoretically, habitat fragmentation could mediate differential dynamics of warm- adapted versus cold-adapted species either by changing existing habitats' suitability or by limiting dispersal into newly suitable habitats (Newson et al., 2014 [↗](#); Opdam & Wascher, 2004 [↗](#)) (Figure 1 [↗](#)). The overall effect of fragmentation on climatic niche tracking of communities can be decomposed into three different mechanisms, each of which has been proposed and supported with different degrees of evidence. First, fragmented habitats may more weakly buffer macroclimate than intact habitats (Ewers & Banks-Leite, 2013 [↗](#)), resulting in fragmented patches experiencing relatively

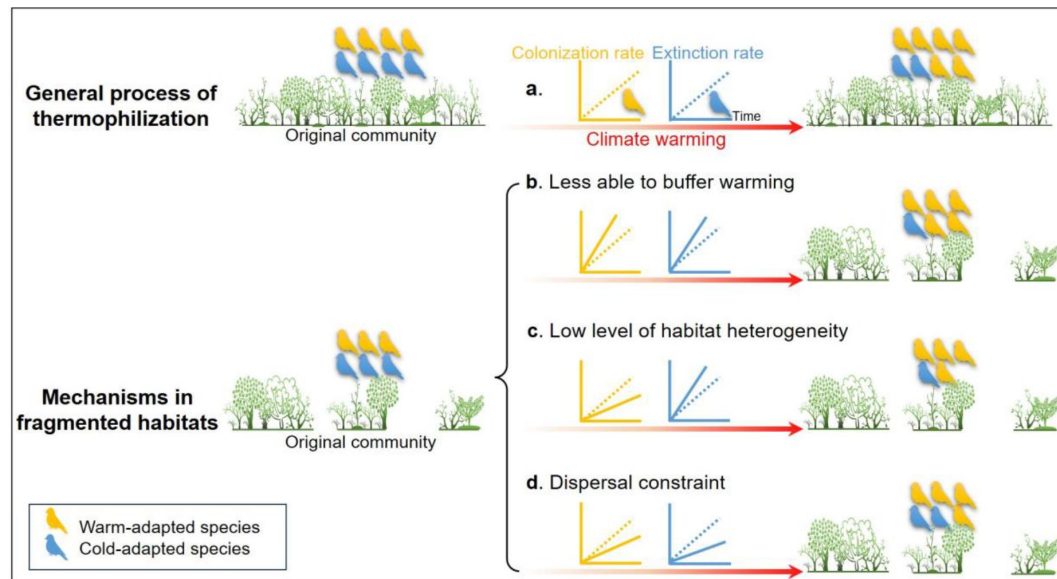


Figure 1.

The general process of thermophilization and the hypothesized framework of thermophilization in fragmented habitats.

Two general processes of thermophilization are shown on the top: increasing colonization rate of warm-adapted species, and increasing extinction rate of cold-adapted species over time (a). Compared to continuous habitats, a hotter microclimate caused by lower buffering ability on fragmented patches may attract more warm-adapted species to colonize while causing cold-adapted species to extirpate or emigrate faster (b). Loss of cold-adapted species can also be exacerbated when highly fragmented patches harbor lower habitat heterogeneity (e.g. resource, microrefugia), which will also reduce colonization of warm-adapted species (c). Isolated patches due to habitat fragmentation will block warm-adapted species' colonization and cold-adapted species' emigration under warming. Relative species richness is shown by the number of bird silhouettes in the community (d).

higher temperatures. Increased temperature, especially in small patches, may boost the colonization or population growth rate of warm-adapted species but cause higher extinction risk or demographic declines of cold-adapted species (**Figure 1b**). This microclimate mechanism has been proposed as an important factor driving thermophilization in plant communities (Zellweger et al., 2020).

Second, larger patches generally contain higher habitat heterogeneity and therefore also encompass higher food availability and microrefugia (Liu et al., 2020). A diversity of habitats and resources is closely related to a species' successful establishment and can act as a buffer against extinction under climate change (Luoto & Heikkinen, 2008). For instance, higher habitat diversity can boost the ability of niche tracking in bird communities (Gäüzère et al., 2017), mainly by fostering species' dispersal into new, climatically suitable sites.

Similarly, in areas where topography generates greater microclimate variation, population losses of plants and insects due to climate change are substantially reduced (Suggitt et al., 2018). We thus expect that the reduction in habitat heterogeneity associated with higher fragmentation will intensify the loss of cold-adapted species and weaken the colonization of warm-adapted species (**Figure 1c**).

Third, habitat fragmentation could prevent or interrupt species' dispersal which is essential for them to track suitable climatic niches (McGuire et al., 2016; Opdam & Wascher, 2004; Thomas et al., 2006). This limitation may come from higher mortality rate with increasing dispersal distance during searching for suitable patches (Brooker et al., 1999), and will influence the colonization-extinction dynamics underlying climate response (Fourcade et al., 2021; Hylander & Ehrlén, 2013). For example, habitat fragmentation makes it too isolated to colonize, causing sedentary butterflies lagged behind climate warming in Britain more severe than mobile ones (Warren et al., 2001). All things being equal, less isolated patches should experience higher increasing rate in colonization of warm-adapted species under warming, benefit from nearer species sources; meanwhile, cold-adapted species should experience higher extinction rate on less isolated patches because they can emigrate easily from these patches under warming (**Figure 1d**).

Although the overall effect of habitat fragmentation on climate-induced community dynamics has been established, few empirical studies have ever considered these multiple mechanisms simultaneously. Here, we use 10 years of bird community data in a subtropical land-bridge island system (Thousand Island Lake, China, **Figure 2**) during a period of consistent climatic warming (**Appendix 1—figure 1**) to explore whether and how community thermal niche composition responds to climate change and how habitat fragmentation mediates the process. This reservoir island system was formed 65 years ago due to dam construction. Specifically, we focused on the following three predictions:

1. We predict the existence of thermophilization of bird communities in our fragmented land-bridge island system, characterized by an increasing temporal trend of CTI.
2. We predict that thermophilization of bird communities should result from both increasing warm-adapted species and abundances and decreasing cold-adapted species and abundances. We also predict that, over time with a warming climate, warm-adapted species should increase their colonization rate while cold-adapted species should increase their extinction rate (**Figure 1a**).

3. We predict that habitat fragmentation – measured as island area and distance to the mainland – will mediate the two main driving processes of thermophilization: colonization of warm-adapted species and extinction of cold-adapted species. Following our theoretical framework, if the climate buffering effect dominates (**Figure 1b**), area effects, we expect the differential trends of warm-adapted species versus cold-adapted species to be more extreme on smaller islands (i.e., even faster colonization of warm-adapted, even faster extinction of cold-adapted) due to weaker microclimate buffering. Alternatively, if the effects of habitat heterogeneity dominate (**Figure 1c**), area effects, cold-adapted species will again show greater extinction, but warm-adapted species will alternatively experience lower colonization on smaller islands (relative to larger islands) due to less abundant resources (Si et al., 2017). Finally, we predict that on more isolated islands (**Figure 1d**), the colonization of warm-adapted species and the extinction of cold-adapted species will increase more slowly because isolation hinders both immigration and emigration.

2 Results

The number of species detected in surveys on each island across the study period averaged 13.36 ± 6.26 species, ranging from 2 to 40 species, with an observed gamma diversity of 60 species. The STI of all 60 birds averaged 19.94 ± 3.58 °C and ranged from 9.30 °C (*Cuculus canorus*) to 27.20 °C (*Prinia inornata*) (**Appendix 1—figure 2**; **Appendix 1—figure 3**). STI between resident species ($n = 47$) and summer visitors ($n = 13$) did not show a significant difference (t-test: $t = 0.23$, $df = 17.82$, $p = 0.82$). No significant correlation was found between STI and species' ecological traits (Pearson correlation $r < \pm 0.22$).

2.1 Thermophilization of bird communities

At the landscape scale, considering species detected across the study area, CTI_{occur} showed no trend (posterior mean temporal trend = 0.414; 95% CrI: -12.751, 13.554) but CTI_{abun} showed a significant increasing trend (temporal trend [mean $\pm$ SE] = 0.327 ± 0.041 , $t = 7.989$, $p < 0.0001$). When measuring CTI trends for individual islands, we found significant increases in CTI for both occurrence- (mean temporal trend = 0.124; 95% CrI: 0.108, 0.137) and abundance-based indices (temporal trend = 0.320 ± 0.021 , $t_{493} = 15.534$, $p < 0.0001$; **Figure 3**).

2.2 Mechanisms underlying CTI trends

Comparing across species with different thermal affinities, we found a weak positive linear relationship between STI and species-specific temporal occupancy trends (t-test: slope = 0.206, $t_{58} = 1.766$, $p = 0.082$; **Figure 4a**), indicating warm-adapted species were marginally more likely to increase in occurrence over time. Decomposing occurrence change to dynamic parameters; however, we found a significant positive relationship between STI and species' temporal trends in colonization (t-test: slope = 0.259, $p = 0.043$) and a significant negative relationship between STI and species' temporal trends in extinction (t-test: slope = -0.414, $p = 0.015$; **Appendix 1—figure 4**). Thus, warm-adapted species generally increased in occupancy over time, which was driven by an increase in colonization probability and a decrease in extinction probability; in contrast, cold-adapted species generally decreased in occupancy over time, which was driven by an increase in extinction probability and a decrease in colonization probability (**Figure 4b**). Considering all species, the influence of a thermal-association gradient underlying occurrence dynamics resulted in a particularly strong correlation between species' trends in colonization and extinction probabilities (Pearson correlation $r = -0.77$; **Figure 4b**).

Similar patterns were found for abundance, where we detected a strong positive relationship (t-test: slope = 0.153, $t_{58} = 3.047$, $p = 0.003$) between STI and each species' temporal trend in abundance: warm-adapted species generally increased abundance while cold-adapted species

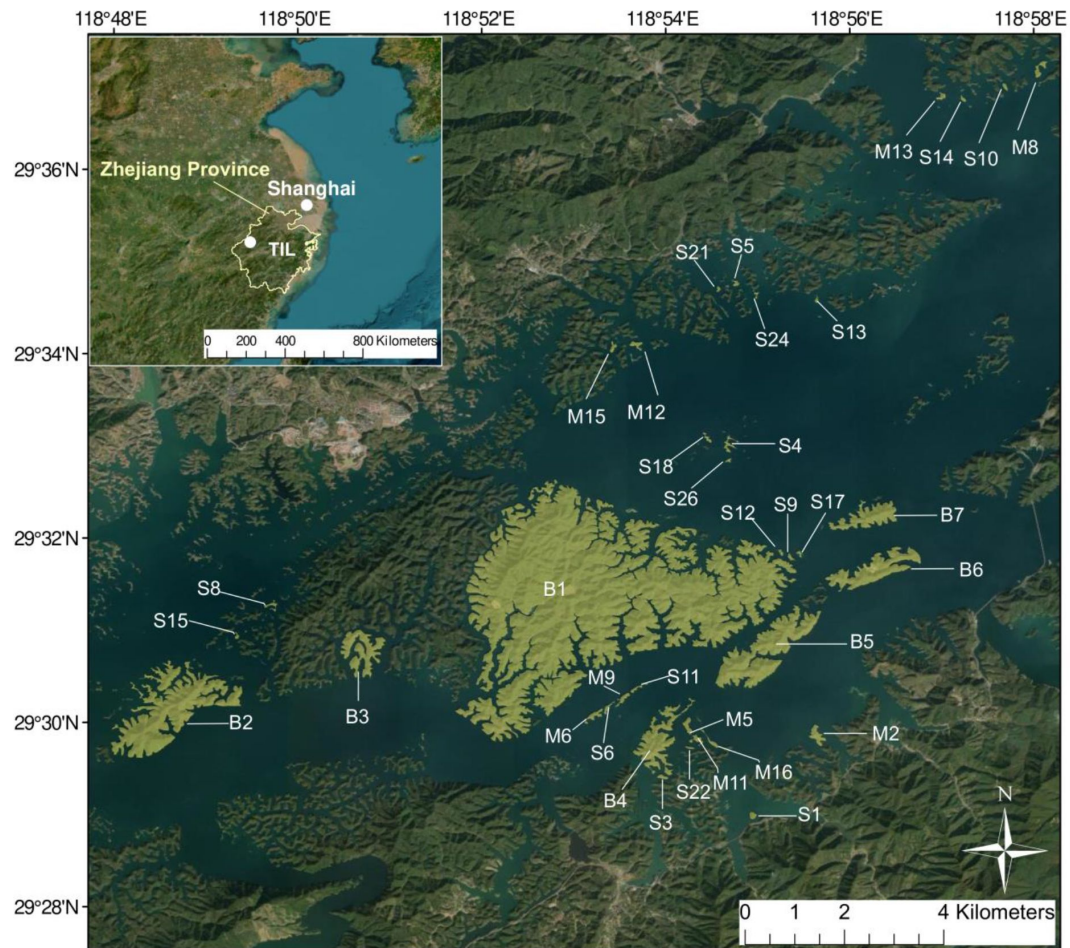


Figure 2.

The location of Thousand Island Lake and the experimental islands.

The map was created using ESRI (Environmental Systems Resource Institute) ArcMap software (version 10.3). The base map sources include Esri, Maxar, Earthstar Geographics, and the GIS User Community.

Figure 3.

Temporal trend in CTI_{occur} and CTI_{abun} in the TIL.

Vertical lines in **(a)** are posterior estimated CTI (mean $\pm$ SD); solid black line and shaded area are predictions and 95% credible intervals extracted from JAGS modeling posterior mean CTI_{occur} as a function of year, and using island identity as a random effect while accounting for the variation in posterior CTI_{occur}. Points and error bars in **(b)** are observed CTI_{abun} (mean $\pm$ 2SD); solid black line and shaded area are predicted values and 95% confidence intervals estimated from LMM modeling observed CTI_{abun} as a function of year, and using island identity as a random effect.

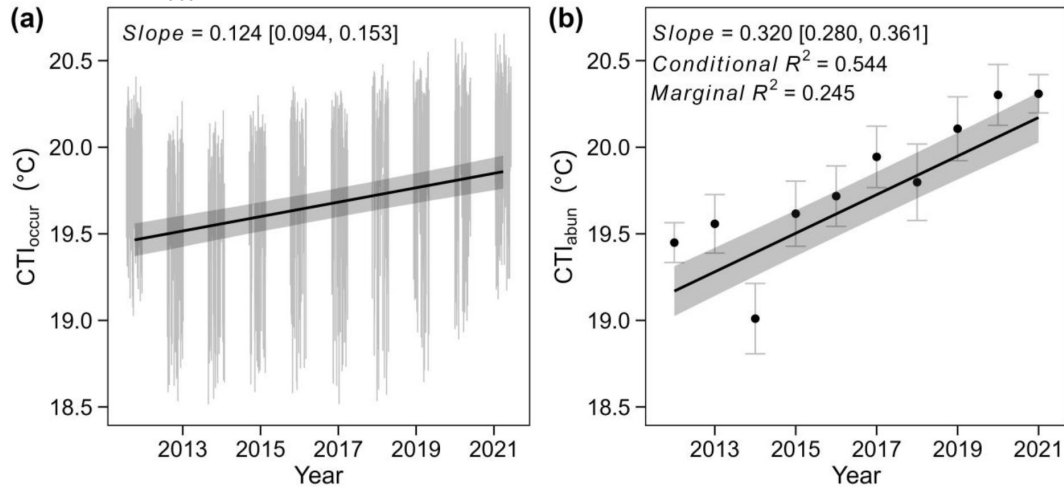
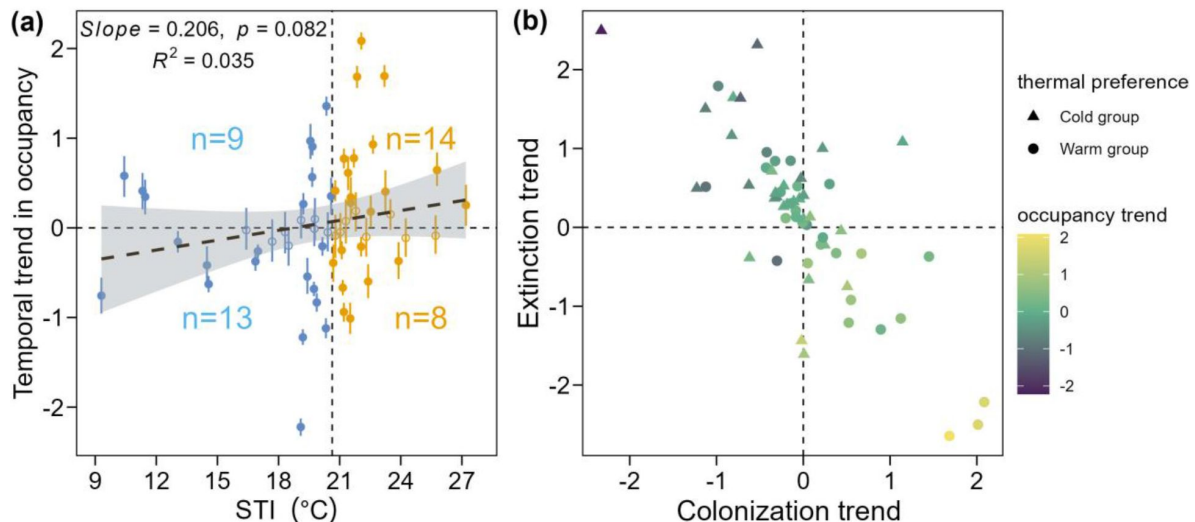


Figure 4.

Relationship between STI and occupancy trend and the relationship between species thermal preference and trend in occupancy, colonization, and extinction rates.

Each point and error bar in **(a)** represents the temporal trend of occupancy rate (posterior mean and 95% credible interval of year effect on occupancy rate) for each species, with the filled dots indicating a significant year effect while the hollow dots indicating nonsignificant year effect. The number of species with significant occupancy trends in each quadrant was added to the plot. The black dashed line and shaded area are the predicted values and 95% confidence intervals of the weighted linear regression model. Each point in **(b)** represents the posterior mean estimate of the colonization trend and extinction trend for each species. The color of the point indicates the temporal trend in occupancy.



decreased over 10 years (**Figure 5**). At the island scale, both total abundance and total species richness increased significantly for the warm-adapted species group while both responses significantly decreased for the cold-adapted species (richness trend, $p < 0.0001$; total abundance trend, $p < 0.0001$; **Appendix 1—figure 5**).

2.3 The mediating effects of fragmentation on colonization and extinction processes

Both colonization and extinction probabilities showed evidence of being moderated by fragmentation (**Figure 6**). Of particular interest was how the temporal trend in dynamic parameters varied as a function of area or isolation (i.e., Year:Area or Year:Isolation effects). Although species-specific interaction terms were not generally significant for most species, overall trends could be assessed across species. In particular, the effect of isolation on temporal dynamics of thermophilization was relatively consistent across cold- (**Figure 6a**) and warm-adapted species (**Figure 6b**); specifically, on islands nearer to the mainland, warm-adapted species (15 out of 15 investigated species) increased their colonization probability at a higher rate over time, while most cold-adapted species (21 out of 23 species) increased their extinction probability at a higher rate.

Island area, in contrast, had a more mixed role in moderating colonization/extinction rates across species. For most warm-adapted species (11 out of 15 investigated species; **Figure 6b**), colonization rates increased faster over time on smaller islands (i.e., negative area:year interaction term); however, the colonization rates of the other four species increased faster on larger islands, including the only significant positive interactive effect (*Dendrocitta formosae*). For cold-adapted species, extinction rates were approximately equally split between increasing faster on smaller islands (12 out of 23 species) and increasing faster on larger islands (11 out of 23 species), including two species, *Spizixos semitorques* and *Pericrocotus cantonensis*, which showed significantly increasing rates of extinction on larger islands.

3 Discussion

Our study explored the driving processes underlying a decade of thermophilization of a bird community breeding in a model fragmented system, specifically testing whether and how habitat fragmentation may mediate the colonization-extinction dynamics which underly shifts in mean community traits. Our results confirm that community thermal niche composition was changing directionally, mainly through gains in warm-adapted species and individuals and losses in cold-adapted species and individuals. We further found that habitat fragmentation influences two processes of thermophilization: colonization rates of warm-adapted species increased faster on smaller and less isolated islands, while the loss rates of most cold-adapted species tended to be exacerbated on less isolated islands.

3.1 Thermophilization of bird communities

Although thermophilization has been found in many taxa (Lajeunesse & Fourcade, 2022) and is more apparent in the tropics than in temperate zones (Freeman et al., 2021), there are few empirical demonstrations of this process in subtropical fragmented systems. We found decadal-scale thermophilization evident in both occurrence- and abundance-based metrics, although the response was considerably stronger for abundance. This result is consistent with work in other systems, where abundance-weighted CTI generally responds more quickly than occurrence-based CTI given climatic warming (Devictor et al., 2008; Lindström et al., 2013; Oliver et al., 2017; Tayleur et al., 2016). Notably, when tested on the landscape scale (versus on individual island

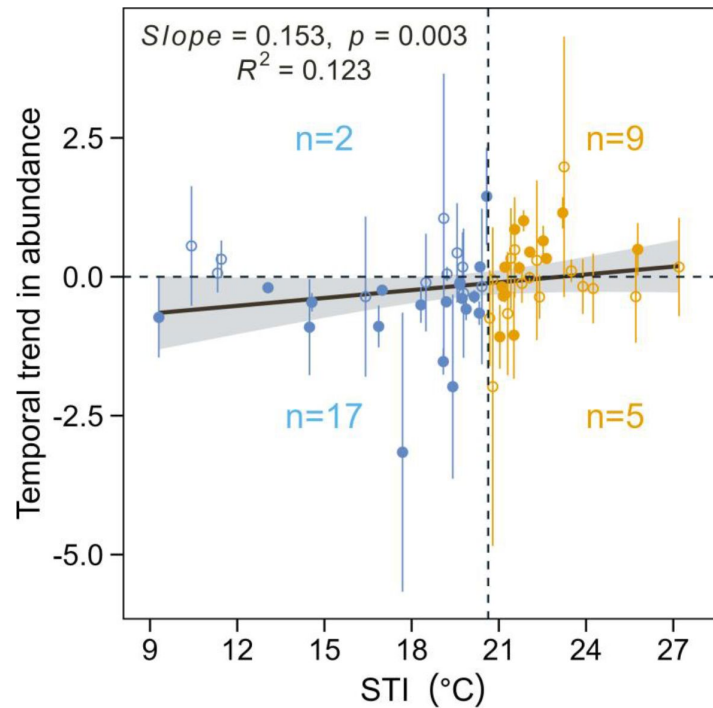


Figure 5.

Relationship between STI and abundance trend.

Each point and error bar represents the temporal trend of abundance (year effect and 95% confidence interval of year effect on occupancy rate) for each species, with the filled dots indicating significant year effect while the hollow dot indicating nonsignificant year effect. Cold-adapted species are plotted in blue and warm-adapted species are plotted in orange. The number in each quadrant indicates the number of species with significant occupancy trends. The black solid line and shaded area are the predicted values and 95% confidence intervals of the weighted linear regression model.

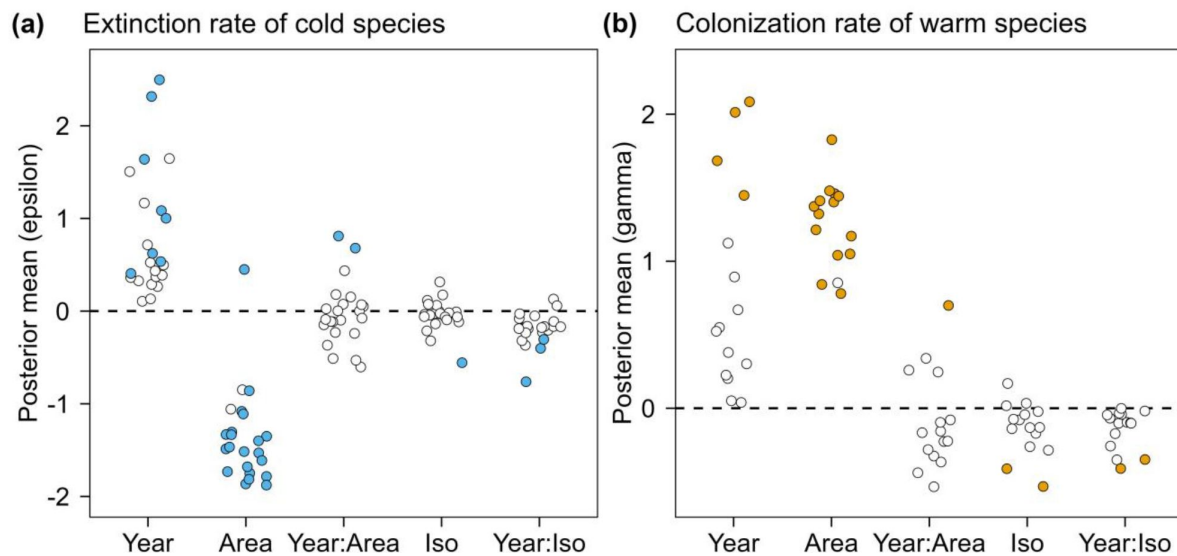


Figure 6.

Posterior estimates related to cold-adapted species' extinction rate and warm- adapted species' colonization rate.

Points are species-specific posterior mean. Only cold- adapted species with positive trends in extinction **(a)** and warm- adapted species with positive trends in colonization **(b)** are shown, which are two main processes contributing positively to thermophilization. Points in blue indicate significant effects (95% or 80% credible intervals) and points in white indicate non-significant effects. All points were jittered slightly by 0.22 in width.

communities), only the abundance-based thermophilization trend was significant, indicating thermophilization of islands was mostly due to inter-island occurrence dynamics, rather than exogenous community turnover.

3.2 Mechanisms underlying CTI trends

Consistent with our expectation, we found an overall increase in occupancy and abundance of warm-adapted species and a decrease in occupancy and abundance of cold-adapted species, which together contributed to thermophilization. This pattern is consistent with many previous studies conducted in continuous habitats in Europe or North America both for breeding and non-breeding bird communities (Curley et al., 2022 [DOI](#); Prince & Zuckerberg, 2015 [DOI](#); Tayleur et al., 2016 [DOI](#)). Also consistent with our expectation was that occupancy change was the product of temporal trends in both colonization and extinction dynamics; namely, warm-adapted species showed increasing colonization and decreasing extinction rates, while, cold-adapted species showed decreasing colonization and increasing extinction rates.

Following that STI is predictably related to temporal trends in colonization and extinction rate (Figure 4 [DOI](#)), we can conclude that STI is an effective broad indicator of relative climatic vulnerability within a community, where species with relatively higher thermal preferences are generally poised to benefit from local climate warming and vice versa. This conclusion is supported by studies that have established relationships between STI and species' population trend (Pearce-Higgins et al., 2015 [DOI](#); Rigal et al., 2023 [DOI](#)), species' contribution analysis (Tayleur et al., 2016 [DOI](#)), or spatial occupancy (Anderson et al., 2023 [DOI](#)). Decomposing average community trait shifts into colonization and extinction dynamics of individual species and their associated traits provides a comprehensive and more nuanced insight into community response to climate change.

There are numerous reasons why we would expect temperature to strongly drive community change, as temperature is related to species' survival through food resources, foraging time, and thermoregulatory requirements (Dunn & Møller, 2019 [DOI](#)). The local population growth of warm-adapted species may mirror their range expansion under climate warming. For example, thermophilization of avian communities in North America is mainly owed to the species that have increased in both range and population size (Curley et al., 2022 [DOI](#)). Contemporary latitudinal range shifts are thought to be primarily resultant of the expansion of polar margins of equatorial-distributed species (La Sorte & Thompson, 2007 [DOI](#); Thomas & Lennon, 1999 [DOI](#)), possibly due to extinction lags at equatorial margins (La Sorte & Jetz, 2012 [DOI](#)). TIL is near the cold edge of most warm-adapted species like *Hemixos castanonotus*, *Pericrocotus solaris*, and *Lonchura punctulata*. Combined with the prediction that most birds in China should move upslope or poleward under climate change (Hu et al., 2020 [DOI](#)), we infer that local climate will continue to shift these warm-adapted species toward a thermal optimum, predicting continued population growth in the near future for these species (Sagarin & Gaines, 2002 [DOI](#)).

Meanwhile, most cold-adapted species in our system demonstrated increasing extinction rates over time. Extinctions are more difficult to detect than range expansions, partly due to temporal lags (Thomas et al., 2006 [DOI](#); Thomas & Lennon, 1999 [DOI](#)) but also due to pervasive bias of false absences in time series data (Tingley & Beissinger, 2009 [DOI](#)). Moreover, species' extinctions are more likely to be mediated by factors such as rainfall, species interactions, and changes in vegetation (Paquette & Hargreaves, 2021 [DOI](#); Thomas & Lennon, 1999 [DOI](#)), which will further lead to unequal rates of extinction relative to colonization. In our community, although there are many cold-adapted species – such as *Streptopelia orientalis*,

Phasianus colchicus, *Passer montanus*, and *Emberiza cioides* – that have northern distributional limits much farther northern than our site, our site is not close to the current southern boundary for these species either. Even so, they still showed considerable loss in TIL, suggesting fragmented landscape exacerbated the negative effect of warming on these cold-adapted species.

Despite the general trend, not all warm-adapted species increased (in abundance or occupancy) and not all cold-adapted species decreased. In other words, not all species contributed equally to the process of thermophilization, as has been seen elsewhere (Tayleur et al., 2016 [↗](#)). Variation in species' response relates to their true thermal tolerance in the study region, which would require additional physiological data to measure and could also be influenced by habitat usage and biotic interactions.

3.3 The mediating effect of fragmentation on colonization and extinction processes

It has long been thought that habitat fragmentation can impact species' climate tracking (Opdam & Wascher, 2004 [↗](#)) but there are only a few empirical studies to date (Fourcade et al., 2021 [↗](#); Warren et al., 2001 [↗](#)). Our research examining colonization-extinction dynamics provides empirical evidence of the possible mechanisms in a land-bridge island system.

As expected, the mediating effect of isolation on the two main processes of thermophilization was relatively consistent, with warm-adapted species colonizing faster and cold-adapted species being extirpated faster on islands nearer to the mainland. This indicates that isolation from population sources can limit the dispersal that underlies range shifts, supporting the general perception that habitat fragmentation is thought to impede climate-driven range expansions (Opdam & Wascher, 2004 [↗](#)) or even block them (Warren et al., 2001 [↗](#)). While we cannot truly distinguish in our system between extinction and emigration, we suspect that given two islands equal except in isolation, if both lose suitability due to climate change, individuals can easily emigrate from the island nearer to the mainland, while individuals on the more isolated island would be more likely to be trapped in place until the species went locally extinct due to a lack of rescue.

Island size also had pervasive effects on underlying thermophilization dynamics. In our study, a large proportion (11 out of 15) of warm-adapted species increasing in colonization rate and half (12 out of 23) of cold-adapted species increasing in extinction rate were changing more rapidly on smaller islands. Similarly, a lower proportion of habitats enhance the colonization of warm-adapted species under climate change in the butterfly assemblage (Fourcade et al., 2021 [↗](#)). The observed colonization-extinction patterns on smaller islands support the hypothesis that warm-adapted species benefit from the lowered thermal buffering of ambient temperature on smaller islands, while the same mechanism may speed up extinction (or emigration) of cold-adapted species (**Figure 1b** [↗](#)). To investigate this mechanistic hypothesis more thoroughly, using understory air temperature data monitored on 20 islands in the breeding season in 2022, we found that temperature buffering ability indeed was diminished on smaller islands (**Appendix 1—figure 6** [↗](#)), further supporting microclimate as a primary driver of fragmentation's mediation of thermophilization (Zellweger et al., 2020 [↗](#)).

The increased extinction rate of some cold-adapted species on smaller islands is also consistent with the hypothesized impacts of reduced habitat heterogeneity (**Figure 1c** [↗](#)).

Previous studies found that areas with greater amounts of habitat featured lower extinction rates of cold-adapted butterfly species (Fourcade et al., 2021 [↗](#)), possibly by facilitating movement to colder microclimates (Suggitt et al., 2018). We thus suppose that habitat heterogeneity, which increases with island area in our system (Liu et al., 2020 [↗](#)), could also mitigate the loss of these relatively cold-adapted species as expected. Thus, as habitat heterogeneity was only consistent with our observed response of cold-adapted species (**Figure 1c** [↗](#)), while microclimate buffering is consistent with observed responses of both warm- and cold-adapted species (**Figure 1b** [↗](#)), our evidence more consistently supports the latter than the former.

Contrary to any expected mechanisms (**Figure 1**), the extinction rate of some cold-adapted species – such as *Spizixos semitorques* and *Pericrocotus cantonensis* – was increased faster on larger islands. Although post hoc, we speculate that these species face stronger predation pressures on larger islands. For example, predators like snakes and felids may also increase activity under climate warming (DeGregorio et al., 2015), but these predators only exist in our system on the larger islands. Besides, *Garrulax canorus*, which is a warm-adapted species, increased faster on larger islands. This species may depend more on habitat heterogeneity than microclimate alone, including food resources and microrefugia, which is related to faster climate tracking (Gäüzère et al., 2017).

4 Materials and Methods

4.1 Study area and islands selection

The Thousand Island Lake (TIL), located in eastern China, was formed in 1959 following the construction of the Xin'anjiang Dam for hydroelectricity (**Figure 2**). When the lake is at its highest, there are 1078 islands with an area larger than 0.25 ha. Currently, about 90% of the forested areas are dominated by Masson pine (*Pinus massoniana*) in the canopy and broad-leaved species (e.g. *Loropetalum chinensei*, *Vaccinium carlesii* and *Rhododendron simsii*) in the sub-canopy and understory (Liu et al., 2020). The climatic zone is a typical monsoon climate. The precipitation is mainly concentrated between April and June with an average yearly rainfall of 1430 mm. The average annual temperature is 17°C (hottest from June to August) and the average daily temperature ranges from −7.6 °C in January to 41.8 °C in July (Si et al., 2017).

For 36 selected islands, we calculated island area and isolation (measured in the nearest Euclidean distance to the mainland) to represent the degree of habitat fragmentation (**Figure 2**). Distance to the mainland is the best distance-based measure in explaining species' turnover in TIL (Si et al., 2014). Since lake formation, the islands have been protected with no logging allowed, allowing natural succession pathways to occur.

4.2 Bird data

We established 53 survey transects on 36 islands with the sampling effort on each island roughly proportional to the logarithm of the island area (Schoereder et al., 2004). As a result, 53 transects on 36 islands were sampled, including eight transect trails on the largest study island (area = 1058 ha), four transects on two islands between 100 and 500 ha, two transects on four islands between 10 and 100 ha, and one on each of the remaining small islands (c. 1 ha for most islands) (Si et al., 2018). Breeding bird communities were surveyed on each transect nine times annually (three times per month from April to June) from 2012 to 2021 (Si et al., 2017). In each survey, observers walked along each transect at a constant speed (20 km/h) and recorded all the birds seen or heard on the survey islands. To minimize the bias, the order and direction of each island surveyed were randomized. The relative abundance of a species on a transect was the maximum abundance recorded across the nine annual surveys. We excluded non-breeding species, nocturnal and crepuscular species, high-flying species passing over the islands (e.g., raptors, swallows) and strongly water-associated birds (e.g., cormorants). Thus, 60 breeding bird species were incorporated into the analysis.

4.3 Climate data

We obtained climate data (monthly average temperature from 2012 to 2021) from the Meteorological Bureau of Chun'an County in Zhejiang Province, China. We calculated breeding season temperature as the average of the mean monthly temperature from April to June in order

to represent the thermal conditions birds experience at this site each year (Lindström et al., 2013). Breeding season temperature increased significantly over 10 years (slope = 0.078, $r^2 = 34.53\%$, $p = 0.04$, **Appendix 1—figure 1**).

4.4 STI and CTI

We followed the methods of Devictor et al. (2008) to calculate a species temperature index (STI) using ArcMap 10.3. STI of a given species is defined as the average temperature of breeding months (April – June) across its distributional range (restricted to the Northern hemisphere) averaged over 1970–2000. Monthly temperature data were obtained from WorldClim at a resolution of 30-arc seconds (<http://www.worldclim.org>). Distributional maps were extracted from Birdlife International 2019 (<http://datazone.birdlife.org/species/requestdis>), in which we selected only the distributional regions where the species is either resident or breeding. All species were divided into two groups indicating their relative thermal preference: species with STI higher than the median STI were labeled as warm-adapted species, and species lower than the median STI were labeled as cold-adapted species (A. E. Bates et al., 2017). The STI of a species can differ depending on distributional and climate data but the rank order of species STI is highly correlated across methods (Barnagaud et al., 2013). We verified the robustness of our STI calculations using different distributional ranges and annual time windows (**Appendix 1—figure 2**).

CTI is a community-level index representing the average species thermal niche of all species or individuals in the community (Devictor et al., 2008). Accordingly, for each community in each year, CTI was calculated as the average STI of all occurring species (hereafter: CTI_{occur}) or counted individuals (hereafter: CTI_{abun}):

$$CTI_{occur,i,t} = \frac{\sum_{n=1}^{N_{i,t}} STI_n}{n}$$

$$CTI_{abun,j,t} = \frac{\sum_{i=1}^{N_{j,t}} (STI_i \times A_{i,j,t})}{\sum_{i=1}^{N_{j,t}} A_{i,j,t}}$$

where $N_{j,t}$ is the total number of species surveyed in the community i in year t , $A_{i,j,t}$ denoted the maximum abundance among nine surveys of the i th species in community j in year t . CTI_{occur} was calculated using occurrence data corrected for imperfect detection (see next section), while CTI_{abun} was estimated using the maximum annual count across 9 surveys. Given that our survey effort was so high (i.e., 9 repeat surveys per year), true abundance should be highly correlated with maximum observed abundance (MacKenzie et al., 2006). We also used these occurrence and abundance metrics from all islands to compute each year's regional CTI at the landscape level.

4.5 Multispecies dynamic occupancy model

Presence-absence data can be highly sensitive to false negatives, so to explore the driving processes of change in CTI_{occur}, and to explore how habitat fragmentation mediated the main processes while accounting for the imperfect detection, we developed a spatially hierarchical dynamic multi-species occupancy model (MSOM) in a Bayesian framework based on the model by Royle and Kéry (2007). We denote y_{nijt} as the observation (detected = 1; undetected = 0) for species n (1–60 species) in survey visit j (1–9 surveys) at transect i (1–53 transects) in year t (1–10 years). Observation, y_{nijt} is assumed to be the result of imperfect detection of the true occurrence status, z_{nit} (1 or 0), and is thus modeled as a Bernoulli-distributed variable with a probability of $p_{nijt} \times z_{nit}$ where p_{nijt} is the probability of detection for a given survey along a transect:

$$y_{nijt} \sim \text{Bernoulli}(z_{nit} * p_{nijt})$$

where z_{nit} is assumed to be of fixed presence/absence status for a given species across all j survey intervals within year t . However, given that individual birds move dynamically across territories within a year, during 9 surveys each year, we broadly interpret our estimates of “occupancy” as

“use” (Si et al., 2018 [↗](#)) to relax the closure assumption (MacKenzie et al., 2004 [↗](#)).

Each site has an initial value of occupancy (ψ_{ni1}) which can be given a uniform prior distribution from 0 to 1, and occupancy (ψ_{nit}) in subsequent years ($t > 1$) is determined based on whether sites become colonized or remain occupied through persistence (MacKenzie et al., 2003 [↗](#)):

$$Z_{nit} \sim \text{Bernoulli}(\psi_{nit})$$

$$\psi_{nit} = Z_{ni,t-1}(1 - \varepsilon_{ni,t-1}) + (1 - Z_{ni,t-1})\gamma_{ni,t-1} \text{ for } t > 1$$

where ψ_{nit} , $\varepsilon_{ni,t-1}$, $\gamma_{ni,t-1}$ are transect-level probabilities of occupancy, extinction, and colonization, respectively.

We modeled the probability of detection, p_{nijt} , as a function of the length of the transect, $length_i$ and the ordinal day of year, day_{ijt} . Island identity was included as a random effect to account for the nonindependence of transects within the same islands.

$$\text{logit}(p_{nijt}) = p0_n + p1_n \text{length}_i + p2_n \text{day}_{ijt} + \text{island}_{REni}$$

To explore how colonization rate and extinction rate change across 10 years and how habitat fragmentation mediated these dynamics processes, we modeled site-level extinction and colonization ($\varepsilon_{ni,t}$, $\gamma_{ni,t}$) each as a logit-linear function of 5 covariates: year, island area, isolation, the interaction between year and area and between isolation and year. Year was added as a random slope, thus allowing the temporal trend in the colonization or extinction to vary with island area and isolation:

$$\begin{aligned} \text{logit}(\varepsilon_{ni,t}) &= \alpha0_n + \alpha1_n \text{area}_i + \alpha2_n \text{isolation}_i + (\alpha3_n + RE_i) \text{year}_t + \alpha4_n \text{area}_i \text{year}_t \\ &\quad + \alpha5_n \text{isolation}_i \text{year}_t + \text{island}_{REni} \\ \text{logit}(\gamma_{ni,t}) &= \beta0_n + \beta1_n \text{area}_i + \beta2_n \text{isolation}_i + (\beta3_n + RE_i) \text{year}_t + \beta4_n \text{area}_i \text{year}_t \\ &\quad + \beta5_n \text{isolation}_i \text{year}_t + \text{island}_{REni} \end{aligned}$$

In all cases, continuous covariates were z-transformed (mean of 0 and a standard deviation of 1). There is no strong correlation between island area and isolation (Pearson correlation $r = -0.214$) so these effects can be considered independent. We fit the MSOM with JAGS (Plummer, 2003 [↗](#)) using the package rjags (Plummer, 2023 [↗](#)) in R v4.3.1 (R Core Team, 2023 [↗](#)). We used vague priors (e.g., normal with $\mu = 0$, $\tau = 0.01$). We ran three chains for 60,000 iterations, discarded the first 40,000 as burn-in and thinned by 20, yielding a combined posterior sample of 3000. We extracted 600 iterations across three chains for Z to calculate CTI_{occur} . Convergence was checked visually with trace plots and confirmed with a Gelman–Rubin statistic < 1.1 (Gelman & Rubin, 1992 [↗](#)). Inference on parameters was made using 95% and 80% Bayesian credible intervals. We checked the posterior predictive ability of the model fit by calculating Bayesian p -values ($p = 0.488$, indicating an unbiased estimation of our model) (Gelman et al., 1996 [↗](#)).

4.6 Statistical analysis

4.6.1 Thermophilization of bird communities

To test for occurrence-based thermophilization at the island level, Bayesian linear regression using R2jags (Su & Yajima, 2021 [↗](#)) was used to derive a posterior estimate of the temporal trend in CTI_{occur} while propagating error in the estimation of CTI_{occur} . The continuous variable *year* was incorporated as the only predictor. To account for the nonindependence of data within islands, we included island identity as a random intercept, thus allowing intercepts to vary across islands. We used vague priors (e.g., normal with $\mu = 0$, $\tau = 0.001$) for intercept and year effect. We ran three

chains for 10,000 iterations, discarded the first 5,000 as burn-in and thinned by 5, yielding a combined posterior sample of 3000. We used the function *MCMCsummary* to get posterior intervals of the intercept and year effect, which were then used for plotting.

To test for abundance-based thermophilization, linear mixed effect models (LMM) with restricted maximum likelihood (REML) were used to model temporal trends in CTI_{abun} using function *lme* in R package nlme (Pinheiro et al., 2017 [↗](#)). As with the occurrence analysis, we used the continuous variable *year* as the only fixed effect and island as a random intercept.

To account for the temporal autocorrelation of CTI from the same island through time, we included the first-order correlation structure (e.g. the corAR1 structure) because observations sampled from the nearby years may be more similar.

We also tested whether the overall regional bird composition (i.e. gamma diversity) experienced thermophilization. Bayesian linear regression was used for modeling temporal trends in regional CTI_{occur} . For abundance, a generalized least square (GLS) model was used to test the relationship between CTI_{abun} and continuous variable *year* while accounting for potential temporal autocorrelation. GLS was conducted using *gls* function in nlme R package (Pinheiro et al., 2023 [↗](#)).

4.6.2 Mechanisms underlying CTI trends

To explore the dynamics underlying trends in CTI_{occur} , specifically, whether thermophilization was accompanied by increasing occupancy of warm-adapted species and decreasing occupancy of cold-adapted species, we extracted 600 posterior samples of occupancy rate per species, and summarized ψ_{nt} by its mean and variation (logit-transformed to meet normality). We then modeled each species' temporal trend in occupancy (ψ_{nt}) over time using Bayesian linear regression in R2jags (Su & Yajima, 2021 [↗](#)) to account for propagated uncertainty in ψ_{nt} . Similarly, to decompose the CTI_{abun} trend, we modeled each species' temporal trend in maximum abundance using a generalized linear mixed effect model with Poisson error structure while accounting for potential overdispersion – via an observation-specific random effect (Harrison, 2014 [↗](#)). Weighted linear regression models tested for a relationship between STI and occupancy trend and between STI and abundance trend. This analysis was conducted using function *lmer* in package lme4 (D. Bates et al., 2015 [↗](#)). Temporal correlations between occupancy and colonization or extinction rates were measured with Pearson correlations, using colonization and extinction rates derived directly from the fitted MSOM.

As a complementary analysis, we also investigated the trend in species richness or total abundance of the two thermal preference groups, separately. The total richness or abundance in each group was calculated on each transect and in each year. We then modeled temporal trends of natural logarithm-transformed richness or abundance using LMM with island identity as the random effect. Partial regression plots from LMM were produced using ggeffects R package (Lüdtke, 2018 [↗](#)). For each model, we checked for normality of residuals and computed goodness-of-fit metrics (R^2) including conditional and marginal R^2 (Nakagawa & Schielzeth, 2013 [↗](#)) using performance R package (Lüdtke et al., 2021 [↗](#)). All continuous fixed effects in the models were standardized (mean = 0 and SD = 1) to ease computation and facilitate the interpretation and comparison of coefficients within and across models (Schielzeth, 2010 [↗](#)).

4.6.3 The mediating effects of habitat fragmentation on colonization and extinction processes

An increasing colonization trend of warm-adapted species and increasing extinction trend of cold-adapted species are two main expected processes that cause thermophilization (Fourcade et al., 2021 [↗](#)). To test our third prediction about the mediating effect of habitat fragmentation, we selected warm-adapted species that had an increasing trend in colonization rate (positive year effect in colonization rate) and cold-adapted species that had an increasing extinction rate

(positive year effect in extinction rate). For each of these species, we extracted their posterior effect of year, area, isolation, and the interaction terms from previously fit models. The interaction terms reveal if habitat fragmentation positively or negatively affects either colonization or extinction processes.

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Author contributions

Juan Liu: Conceptualization; data curation; formal analysis; methodology; visualization; writing – original draft; writing – review and editing. **Morgan W. Tingley:** formal analysis; methodology; visualization; writing – review and editing. **Qiang Wu:** data curation; visualization; writing – original draft (supporting); writing – review and editing. **Peng Ren:** data curation; visualization; writing – original draft (supporting); writing – review and editing. **Tinghao Jin:** data curation; formal analysis; methodology; visualization; writing – original draft (supporting); writing – review and editing. **Ping Ding:** Supervision; conceptualization; data curation; funding acquisition; methodology; writing – review and editing. **Xingfeng Si:** Supervision; conceptualization; methodology; data curation; funding acquisition; writing – review and editing.

Competing interests

The authors declare no conflicts of interests.

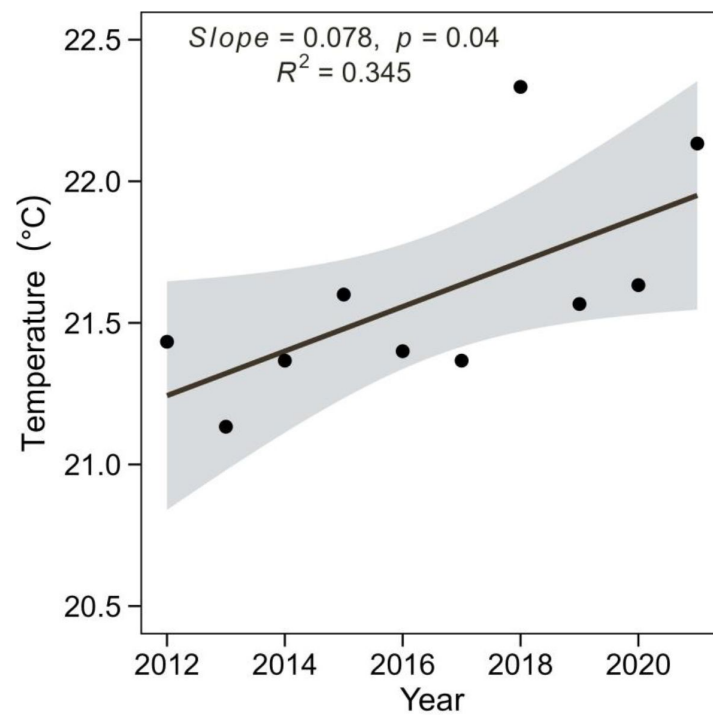
Data availability

The data that support the findings of this study are openly available in Figshare at <https://figshare.com/s/7a16974114262d280ef7> .

Code availability

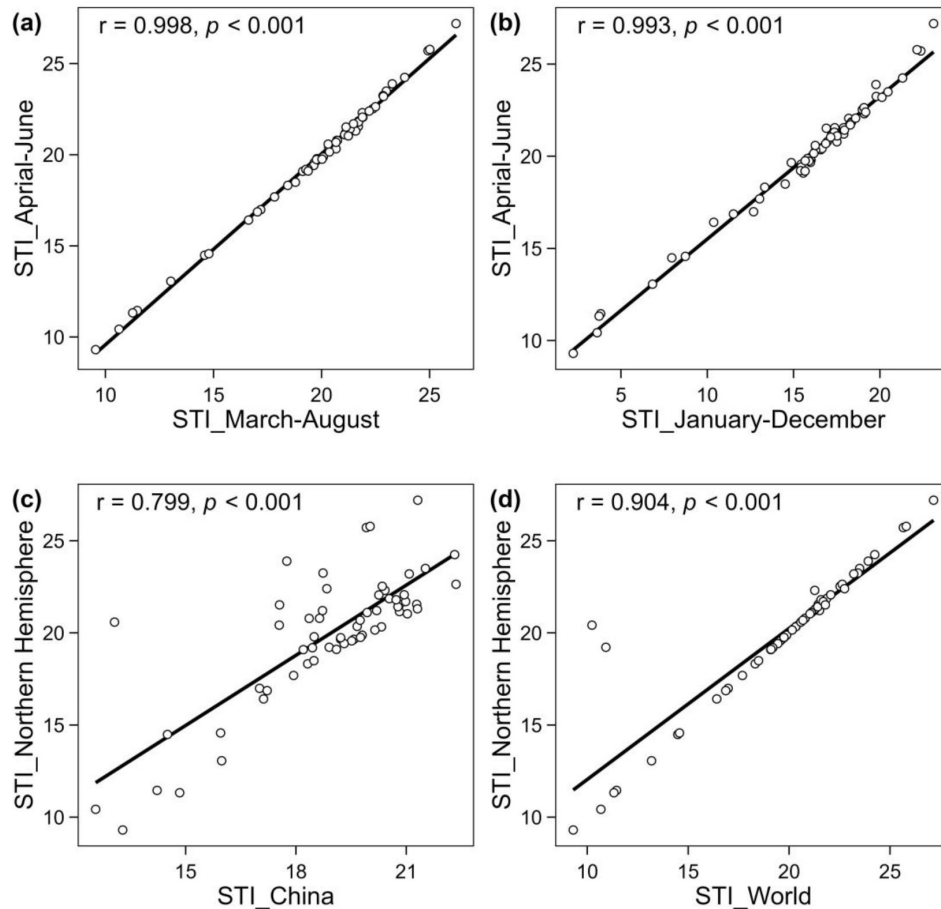
The code that supports the findings of this study is openly available in Figshare at <https://figshare.com/s/7a16974114262d280ef7> .

Appendix 1



Appendix 1—figure 1.

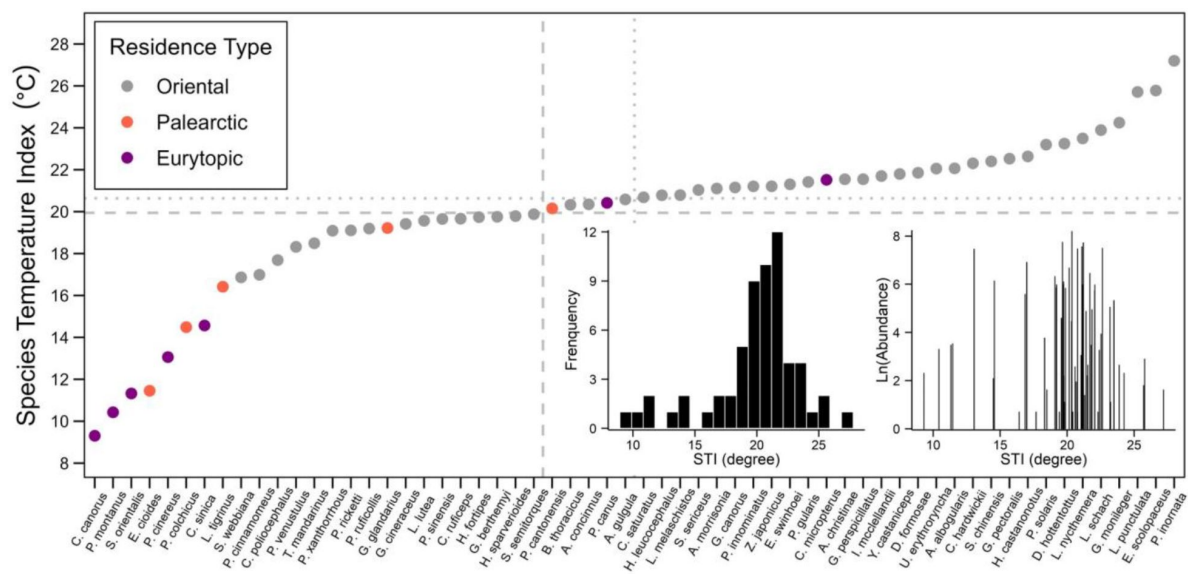
Breeding season temperature from the year 2012 to 2021 in the Thousand Island Lake, China. Breeding season temperature was calculated as the average of the mean monthly temperature for breeding seasons (April – June). Black lines and grey shade indicate the regression line and confidence interval of the linear regression model (slope = 0.078, $r^2 = 34.53\%$, $p = 0.04$).



Appendix 1—figure 2.

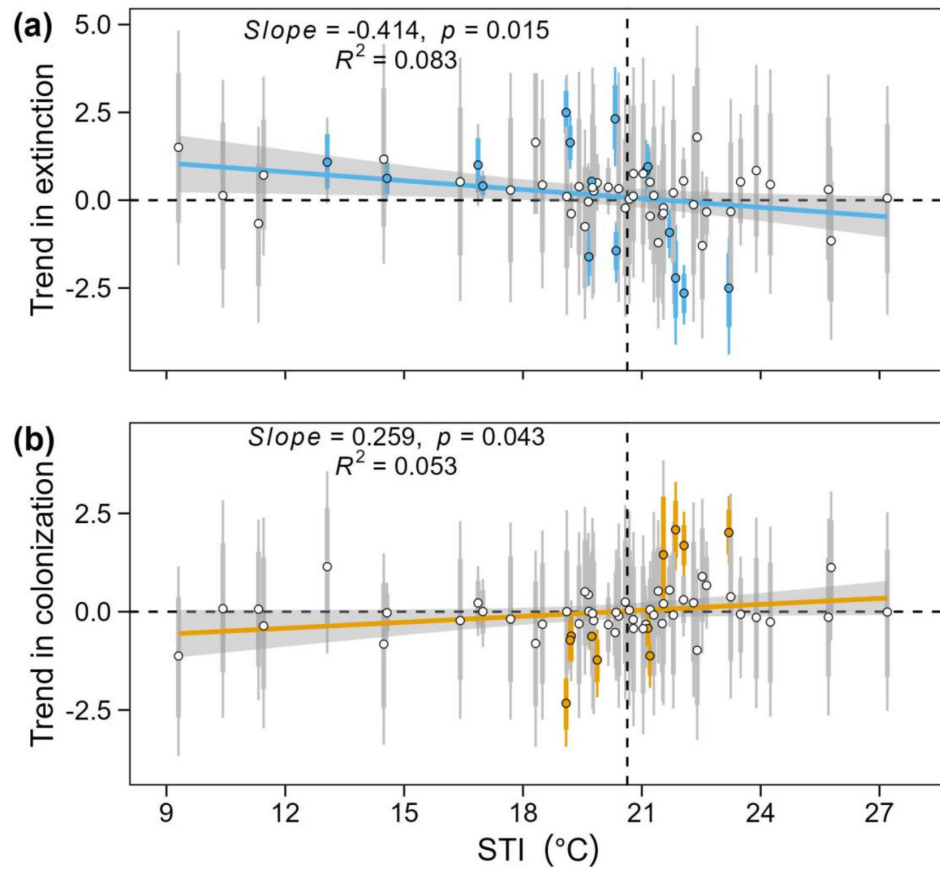
Robustness of STI to changes in data resources.

The units on the horizontal and vertical axes are degrees Celsius. Pearson correlation coefficients between STI used in our study and STI calculated with different time-windows: March to August (a) and January to December (b), were significantly high. Pearson correlation coefficients between STI used in our study and STI calculated with different distributional ranges: the distributional range in China (c) and distributional range across the world (d), were significantly high.



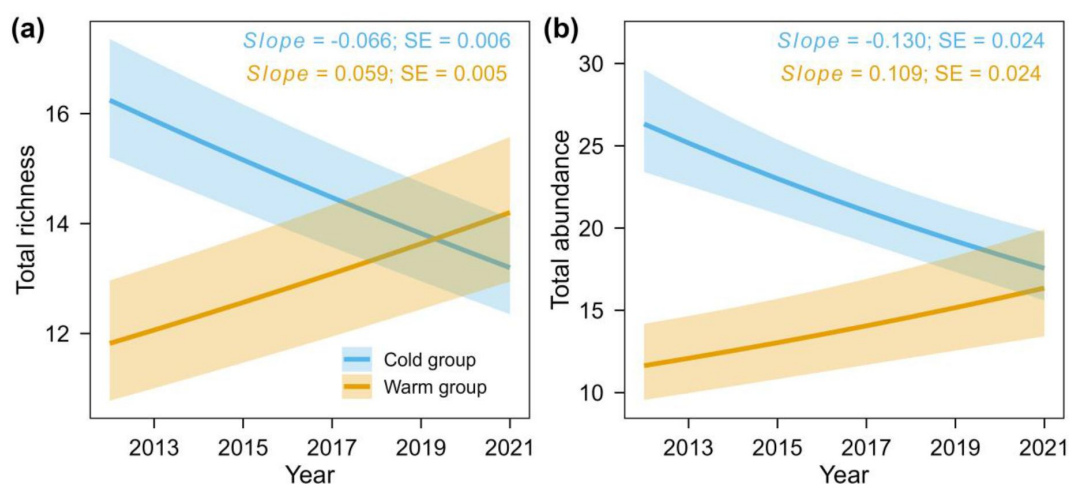
Appendix 1—figure 3.

Species Temperature Index of 60 species in the Thousand Islands Lake, China. The dotted vertical line indicates the median of STI values; the dashed vertical line indicates the mean of STI values. Grey points are Oriental species, red points are Palearctic species and purple points are Eurytopic species. The bottom right presents the frequency of STI values (right) and the overview relationship between total abundance (natural logarithm transversion of total abundance over 10 years and on all islands) and STI.



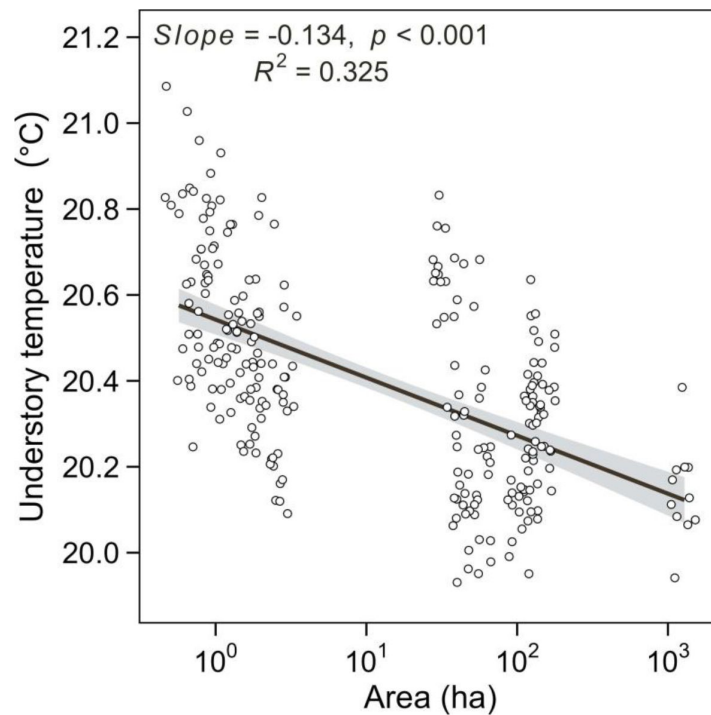
Appendix 1—figure 4.

Relationship between STI and mean estimates and 95%, and 80% credible intervals of posterior distribution for temporal trends of species extinction rate



Appendix 1—figure 5.

(a) and species colonization rate (b). Points and error bars in blue or orange indicate significant posterior estimates and error bars in grey indicate non-significant effects. Results of linear regression models modeling posterior mean of temporal trends in colonization or extinction weighted by standard deviations were added to plots. Partial regression effects of year effect on total species richness **(a)** and total abundance **(b)** for warm versus cold-adapted species groups. The solid lines indicate significant effects. The shaded bands are 95% confidence intervals of fitted values with other variables at their mean values. Total richness was calculated as the mean of the posterior estimate of species absence/presence, while total abundance was calculated from raw maximum abundance data.



Appendix 1—figure 6.

Relationship between island area and understory breeding season air temperature of 20 islands monitored in the year 2022.

The understory air temperature was monitored in 20 selected study islands differing in island area. Each plot represents the average air temperature (April – June) of a site within the islands. Lines and shaded areas are predicted values and 95% confidence intervals from LM models.

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Reviewer #1 (Public Review):

Summary:

This study reports on the thermophilization of bird communities in a network of islands with varying areas and isolation in China. Using data from 10 years of transect surveys, the authors show that warm-adapted species tend to gradually replace cold-adapted species, both in terms of abundance and occurrence. The observed trends in colonisations and extinctions

are related to the respective area and isolation of islands, showing an effect of fragmentation on the process of thermophilization.

Strengths:

Although thermophilization of bird communities has been already reported in different contexts, it is rare that this process can be related to habitat fragmentation, despite the fact that it has been hypothesized for a long time that it could play an important role. This is made possible thanks to a really nice study system in which the construction of a dam has created this incredible Thousand Islands lake. Here, authors do not simply take observed presence-absence as granted and instead develop an ambitious hierarchical dynamic multi-species occupancy model. Moreover, they carefully interpret their results in light of their knowledge of the ecology of the species involved.

Weaknesses:

Despite the clarity of this paper on many aspects, I see a strong weakness in the authors' hypotheses, which obscures the interpretation of their results. Looking at Figure 1, and in many sentences of the text, a strong baseline hypothesis is that thermophilization occurs because of an increasing colonisation rate of warm-adapted species and extinction rate of cold-adapted species. However, there does not need to be a temporal trend! Any warm-adapted species that colonizes a site has a positive net effect on CTI; similarly, any cold-adapted species that goes extinct contributes to thermophilization.

Another potential weakness is that fragmentation is not clearly defined. Generally, fragmentation *sensu lato* involves both loss of habitat area and changes in the spatial structure of habitats (i.e. fragmentation *per se*). Here, both area and isolation are considered, which may be slightly confusing for the readers if not properly defined.

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Reviewer #2 (Public Review):

Summary:

This study addresses whether bird community reassembly in time is related to climate change by modelling a widely used metric, the community temperature index (CTI). The authors first computed the temperature index of 60 breeding bird species thanks to distribution atlases and climatic maps, thus obtaining a measure of the species realized thermal niche.

These indices were aggregated at the community level, using 53 survey transects of 36 islands (repeated for 10 years) of the Thousand Islands Lake, eastern China. Any increment of this CTI (i.e. thermophilization) can thus be interpreted as a community reassembly caused by a change in climate conditions (given no confounding correlations).

The authors show thanks to a mix of Bayesian and frequentist mixed effect models to study an increment of CTI at the island level, driven by both extinction (or emigration) of cold-adapted species and colonization of newly adapted warm-adapted species. Less isolated islands displayed higher colonization and extinction rates, confirming that dispersal constraints (created by habitat fragmentation *per se*) on colonization and emigration are the main determinants of thermophilization. The authors also had the opportunity to test for habitat amount (here island size). They show that the lack of microclimatic buffering resulting from less forest amount (a claim backed by understory temperature data) exacerbated the rates of cold-adapted species extinction while fostering the establishment of warm-adapted species.

Overall these findings are important to range studies as they reveal the local change in affinity to the climate of species comprising communities while showing that the habitat fragmentation VS amount distinction is relevant when studying thermophilization. As is, the manuscript lacks a wider perspective about how these results can be fed into conservation biology, but would greatly benefit from it. Indeed, this study shows that in a fragmented reserve context, habitat amount is very important in explaining trends of loss of cold-adapted species, hinting that it may be strategic to prioritize large habitats to conserve such species. Areas of diverse size may act as stepping stones for species shifting range due to climate change, with small islands fostering the establishment of newly adapted warm-adapted species while large islands act as refugia for cold-adapted species. This study also shows that the removal of dispersal constraints with low isolation may help species relocate to the best suitable microclimate in a heterogeneous reserve context.

Strength:

The strength of the study lies in its impressive dataset of bird resurveys, that cover 10 years of continued warming (as evidenced by weather data), 60 species in 36 islands of varying size and isolation, perfect for disentangling habitat fragmentation and habitat amount effects on communities. This distinction allows us to test very different processes mediating thermophilization; island area, linked to microclimatic buffering, explained rates for a variety of species. Dispersal constraints due to fragmentation were harder to detect but confirms that fragmentation does slow down thermophilization processes.

This study is a very good example of how the expected range shift at the biome scale of the species materializes in small fragmented regions. Specifically, the regional dynamics the authors show are analogous to what processes are expected at the trailing and colonizing edge of a shifting range: warmer and more connected places display the fastest turnover rates of community reassembly. The authors also successfully estimated extinction and colonization rates, allowing a more mechanistic understanding of CTI increment, being the product of two processes.

The authors showed that regional diversity and CTI computed only by occurrences do not respond in 10 years of warming, but that finer metrics (abundance-based, or individual islands considered) do respond. This highlights the need to consider a variety of case-specific metrics to address local or regional trends. Figure Appendix 2 is a much-appreciated visualization of the effect of different data sources on Species thermal Index (STI) calculation.

The methods are long and diverse, but they are documented enough so that an experienced user with the use of the provided R script can follow and reproduce them.

Weaknesses:

While the overall message of the paper is supported by data, the claims are not uniformly backed by the analysis. The trends of island-specific thermophilization are very credible (Figure 3), however, the variable nature of bird observations (partly compensated by an impressive number of resurveys) propagate a lot of errors in the estimation of species-specific trends in occupancy, abundance change, and the extinction and colonization rates. This materializes into a weak relationship between STI and their respective occupancy and abundance change trends (Figure 4a, Figure 5, respectively), showing that species do not uniformly contribute to the trend observed in Figure 3. This is further shown by the results presented in Figure 6, which present in my opinion the topical finding of the study. While a lot of species rates response to island areas are significant, the isolation effect on colonization and extinction rates can only be interpreted as a trend as only a few species have a significant effect. The actual effect on the occupancy change rates of species is hard to grasp, and this trend has a potentially low magnitude (see below).

While being well documented, the myriad of statistical methods used by the authors ampere the interpretation of the figure as the posterior mean presented in Figure 4b and Figure 6 needs to be transformed again by a logit-1 and fed into the equation of the respective model to make sense of. I suggest a rewording of the caption to limit its dependence on the method section for interpretation.

By using a broad estimate of the realized thermal niche, a common weakness of thermophilization studies is the inability to capture local adaptation in species' physiological or behavioral response to a rise in temperature. The authors however acknowledge this limitation and provide specific examples of how species ought to evade high temperatures in this study region.

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Reviewer #3 (Public Review):

Summary:

Juan Liu et al. investigated the interplay between habitat fragmentation and climate-driven thermophilization in birds in an island system in China. They used extensive bird monitoring data (9 surveys per year per island) across 36 islands of varying size and isolation from the mainland covering 10 years. The authors use extensive modeling frameworks to test a general increase in the occurrence and abundance of warm-dwelling species and vice versa for cold-dwelling species using the widely used Community Temperature Index (CTI), as well as the relationship between island fragmentation in terms of island area and isolation from the mainland on extinction and colonization rates of cold- and warm-adapted species. They found that indeed there was thermophilization happening during the last 10 years, which was more pronounced for the CTI based on abundances and less clearly for the occurrence-based metric. Generally, the authors show that this is driven by an increased colonization rate of warm-dwelling and an increased extinction rate of cold-dwelling species. Interestingly, they unravel some of the mechanisms behind this dynamic by showing that warm-adapted species increased while cold-dwelling decreased more strongly on smaller islands, which is - according to the authors - due to lowered thermal buffering on smaller islands (which was supported by air temperature monitoring done during the study period on small and large islands). They argue, that the increased extinction rate of cold-adapted species could also be due to lowered habitat heterogeneity on smaller islands. With regards to island isolation, they show that also both thermophilization processes (increase of warm and decrease of cold-adapted species) were stronger on islands closer to the mainland, due to closer sources to species populations of either group on the mainland as compared to limited dispersal (i.e. range shift potential) in more isolated islands.

The conclusions drawn in this study are sound, and mostly well supported by the results. Only a few aspects leave open questions and could quite likely be further supported by the authors themselves thanks to their apparent extensive understanding of the study system.

Strengths:

The study questions and hypotheses are very well aligned with the methods used, ranging from field surveys to extensive modeling frameworks, as well as with the conclusions drawn from the results. The study addresses a complex question on the interplay between habitat fragmentation and climate-driven thermophilization which can naturally be affected by a multitude of additional factors than the ones included here. Nevertheless, the authors use a well-balanced method of simplifying this to the most important factors in question (CTI change, extinction, and colonization, together with habitat fragmentation metrics of isolation

and island area). The interpretation of the results presents interesting mechanisms without being too bold on their findings and by providing important links to the existing literature as well as to additional data and analyses presented in the appendix.

Weaknesses:

The metric of island isolation based on the distance to the mainland seems a bit too oversimplified as in real life the study system rather represents an island network where the islands of different sizes are in varying distances to each other, such that smaller islands can potentially draw from the species pools from near-by larger islands too - rather than just from the mainland. Thus a more holistic network metric of isolation could have been applied or at least discussed for future research. The fact, that the authors did find a signal of island isolation does support their method, but the variation in responses to this metric could hint at a more complex pattern going on in real-life than was assumed for this study. Further, the link between larger areas and higher habitat diversity or heterogeneity could be presented by providing evidence for this relationship. The authors do make a reference to a paper done in the same study system, but a more thorough presentation of it would strengthen this assumption further.

Despite the general clear patterns found in the paper, there were some idiosyncratic responses. Those could be due to a multitude of factors which could be discussed a bit better to inform future research using a similar study design.

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Author response:

We would like to 1) response one comment from the public review, which is also related to the eLife assessment, and 2) give provisional author responses.

(1) Regarding the definition of the colonization-extinction rate, the first reviewer may misunderstand it: “However, there does not need to be a temporal trend! Any warm-adapted species that colonizes a site has a positive net effect on CTI; similarly, any cold-adapted species that goes extinct contributes to thermophilization.” We here clarify the definition:

In a single iteration of our MSOM (Multi-species occupancy model), the occupancy rate of species[n] in transect[i] from year[t-1] to year[t] is related to the colonization rate and extinction rate, and is defined as:

$$\text{muz}[n,i,t] = z[n,i,t-1]*(1-\text{eps}[n,i,t-1]) + (1-z[n,i,t-1])*gam[n,i,t-1],$$
 (also shown in Line411 in our MS).

If the colonization rate (gam) and extinction rate (eps) remain constant, the occupancy rate(muz) will be a constant number which is related to the state of real occupancy (0 or 1). The occupancy rate will only increase if colonization rate increases (or the extinction rate decreases). That is why we are considering the temporal trend in colonization/extinction rate.

(2) Provisional author responses:

We will revise and improve the manuscript according to the public reviews and mainly focus on:

(1) clarify the general definition of habitat fragmentation in the Introduction.

(2) provide a wider perspective about how our results can be applied to conservation biology in the Discussion.

(3) discuss the diversity of isolation metrics for future research and provide more evidence about the link between larger areas and higher habitat diversity or heterogeneity.

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